

eCos For Limitless Configurations Of Your EMBEDDED OS

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When the acronym for software tells you only what it is and nothing else, you can surely expect a no-nonsense approach from it. embedded Configurable operating system, or eCos, is the embedded OS we look at in this month's article.

eCos is a free and open source, real-time OS used in many embedded systems and applications worldwide. It is mostly used to run applications that need a single process but multiple threads.

The design of eCos allows you to customise it to precise application requirements, be it run-time performance or hardware specificity. Implemented in C/C++ , eCos provides you with compatible layers and APIs for POSIX and µTRON.

An embedded OS is different from a normal OS in many ways, most of which are related to the scarcity of resources, need for small form factor and the very application that these are to run. An embedded OS is one that is specifically configured for a certain hardware configuration. 'Do specific tasks efficiently' is the mantra for such OSes. With the Internet of Things (IoT) craze only expected to grow, these OSes will be the driving factor for all application development in this field.

Where does eCos come into all of this

Today, with a bevy of industrial real-time solutions available for processing, there are certain factors that determine which OS to select for resource-constrained applications. eCos sets itself apart by competing on three parameters, namely, cost,

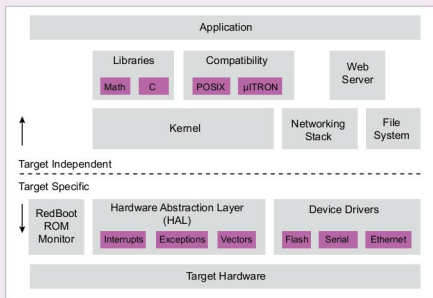


Fig. 1: The eCos layered software architecture

control and technical fit.

Cost. Being totally free helps a lot in this regard when you are basically telling your users that they would not have to pay anything for development-related activities. eCos is portable across chip architectures, which makes it well-suited for software developers of high-volume applications in consumer electronics, telecommunications, automotive and other embedded applications.

Control. The open source nature of eCos lets users upload their projects, modify the source code, help others and, on the whole, be the good samaritan they tell everyone they are.

Technical fit. As mentioned before, the massive configurability options come in handy while using eCos. With more than 200 source-level configuration points, users can bet their last rupee that they will

be able to match their need with the requisite configuration.

What is the relation between eCos and Linux

Not as scandalous as it might sound, so un-brace yourself. Since eCos originates from RedHat, the distributors of Linux, there are certain corners where they believe that there is some serious correlation between Linux and eCos. Though eCos has a separate compatibility level for several Linux APIs, that does not mean that these share common codes.

Codes for Linux and eCos are not only entirely different, these also serve extremely separate purposes. Linux is not inherently real-time, as it does not have a real-time kernel associated with it. When you try to pair it with an external RT kernel, Linux basically just sits idly by or as a separate virtual machine. Hence,